

An evidence broker for Australian schools

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Introduction

Curiosity is at the heart of teaching and learning. This quality of inquisitive thinking, exploration and investigation facilitates intrinsic motivation, prepares the brain for learning and makes subsequent learning more rewarding (Gruber, Gelman and Ranganath, 2014; Stenger, 2014). Great teachers foster curiosity in their students and have it themselves for the knowledge and skills they teach. They also bring that spirit of inquiry to their teaching practice, searching for ways to increase their positive impact (AITSL, 2011).

A culture of constant inquiry to support better learning is essential for improving education. Education is a fundamental driver for the health of our society and economy. It is also the best way to break the stubborn link between a child's vulnerable circumstances and narrowed life chances.¹ Given both the moral imperative and the social good, it should be a shared national priority to help all children make the most of their talents; for themselves, for their families and communities and for Australian society.

Australia needs a shared spirit of inquiry and endeavour now, more than ever, in schooling because:

- there is an overall decline in performance on standardised national and international measures of academic performance;²
- there is growing divide in performance based on family income and background;³ and
- there are lower engagement levels and declining perceptions of relevance reported by students.⁴

These are complex challenges and they cannot be solved through schooling alone. However, in every case schools are fundamental to the solution.

Teachers and school leaders are not alone in seeking good answers to these questions. System leaders search for the right investments and policies to support improved learning, professional networks share knowledge and experiences, and researchers work to find new and better approaches. However, all of their work

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only makes a difference when it actually improves learning relationships in the school and when it changes what teachers and students do in the classroom (Hattie, 2003).

Pragmatically (as school leaders know only too well), these challenges can only be addressed with the human, physical and financial resources that are available. Whilst debate rightly occurs about how much is enough (Doyle, 2017; Johansson, 2016), schools strive to make the best use of what they have. This creates another type of inquiry for leaders and teachers’ – ‘Are we using our resources in ways that generate the best learning for our students?’ This, too, is a complex question; it may change between schools, depending on their context and the specific needs of their community. It even changes within a school at different times.

Our organisation, Evidence for Learning (2017d), aims to help educators find better answers to these questions for themselves and their learners. We recognise schools as the most effective place to focus attention and support. Our mission is helping educators increase learning through better evidence. We strengthen their decision making by making robust and actionable research easily available so that, combined with a spirit of curiosity and their professional expertise, they make better choices for their students.

The role we play is known as an ‘evidence broker’.⁵ Evidence for Learning sits at the intersection of policy, research and practice, supporting all three domains but not privileging any of them.

We are independent, national, cross sectoral and non-profit. Our founding partners are Social Ventures Australia,⁶ the Commonwealth Bank⁷ and the UK’s Education Endowment Foundation,⁸ which is an established evidence broker for the English school system.

Evidence for Learning has been operating for nearly three years and, true to the spirit of inquiry, developed the organisation as a pilot to inform Australia’s education system on how an evidence broker could assist in addressing the complex challenges to better learning for all. In this paper we tell the story of our work and learnings so far.

What works best and how do you know?

The essence of a profession lies in its ability to make decisions (Hargreaves and Fullan, 2012). Leaders draw from the ‘findings from a range of cases in exercising professional judgement’ (Caldwell, 2016, p 7). This decision making is influenced by ‘personal and professional values and beliefs, and affected by wider political and educational contexts, policies and changes’ (Nelson and Campbell, 2017, p 128).

Evidence is critical in the exercise of professional judgement. A broad definition of evidence is ‘the available knowledge and information indicating whether a belief or proposition is true or valid’.⁹ *Empirical evidence*¹⁰ is the bedrock of the scientific method. In the humanities it includes other forms of knowledge to support a position.¹¹ In all fields, it is about establishing confidence in the proposition being put forward. Both the quantity and quality of the evidence make a difference to levels of confidence.

Evidence-informed decisions are about ‘integrating professional expertise with the best external evidence from research to improve the quality of practice’ (Sharples, 2013, p 7). leading to what might be characterised as *evidence-based practice*. Crucially, this is not about ‘prescribing what goes on from a position of unchallenged authority’ (Sharples, 2013, p 7). Central to the work of Evidence for Learning is the view that school leaders and teachers

are ‘active inquirers’ who are ‘bound together by norms and structures akin to a scientific community’ (Bryk, 2015, p 469). It is educators who create ‘practice-based evidence’ that is ‘evidence to help us discern whether any specific changes attempted are actually improvements’ (Bryk, 2015, p 475).

Not all evidence is equal. How should the classroom teacher distinguish fads and marketing hype from robust evidence based in practice? One model to help the ‘active inquirer’ to determine whether a program or practice is likely to assist them in the challenge they face is a hierarchy of evidence shown in Figure 1.

This pyramid shows how different kinds of evidence can be critically assessed, moving from personal experience and the anecdotal at Level 4, through to case studies with defined measures and results at Level 3, up to experimental studies testing for causation at Level 2 and finally to repeatable findings from multiple studies at Level 1.

The benefit of the hierarchy is to recognise that even the best innovations begin with limited evidence of impact which are then

supported and validated (or discounted) over time with increasing degrees of rigour and confidence.

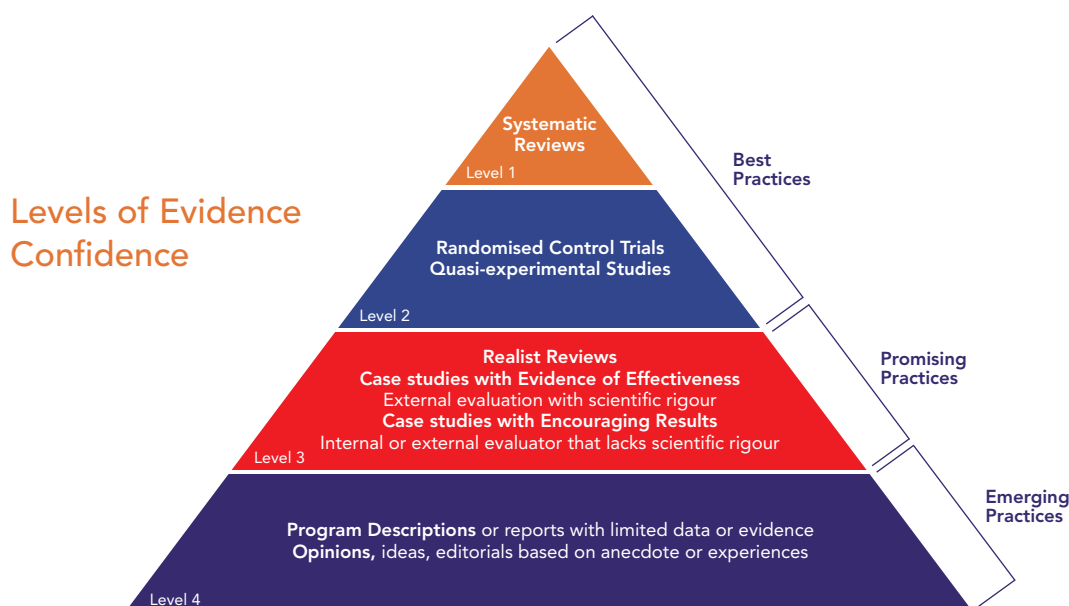
There are two important caveats to this kind of model, both of which highlight the eternal importance of professional judgement.

Firstly, even Level 1 Systematic Reviews are not evidence for ‘what works’ but rather ‘what worked’ – at another time in another place or places for other learners. Whilst there might be greater confidence of a likely positive effect (given the number of studies in which the findings have been repeated), it is not a guarantee of success in every circumstance.

Secondly, let us not just ask ‘what worked’ but ‘what worked, for whom, under what conditions’.¹² There may be prerequisites (such as teacher knowledge or amount of teaching time) to get the same effect at another time and place. Or the approach may work better for some learners than others (such as learners who have already mastered basic knowledge). It is essential to understand these nuances from the evidence generated to properly inform professional decision making.

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Figure 1. Hierarchy of evidence



An evidence ecosystem

With our core focus on helping educators make better choices with evidence, Evidence for Learning created an ecosystem model for thinking about the creation of *practice-based* evidence in schools that drives the generation of further knowledge to support *evidence-based* practice.

The evidence ecosystem has two cycles, one focused on a process in the school (Impact Evaluation Cycle) and the other looking at a process from those outside the school (Wider Evidence Chain). The two are intimately connected and reinforcing processes, as shown in Figure 2.

The impact evaluation cycle in schools

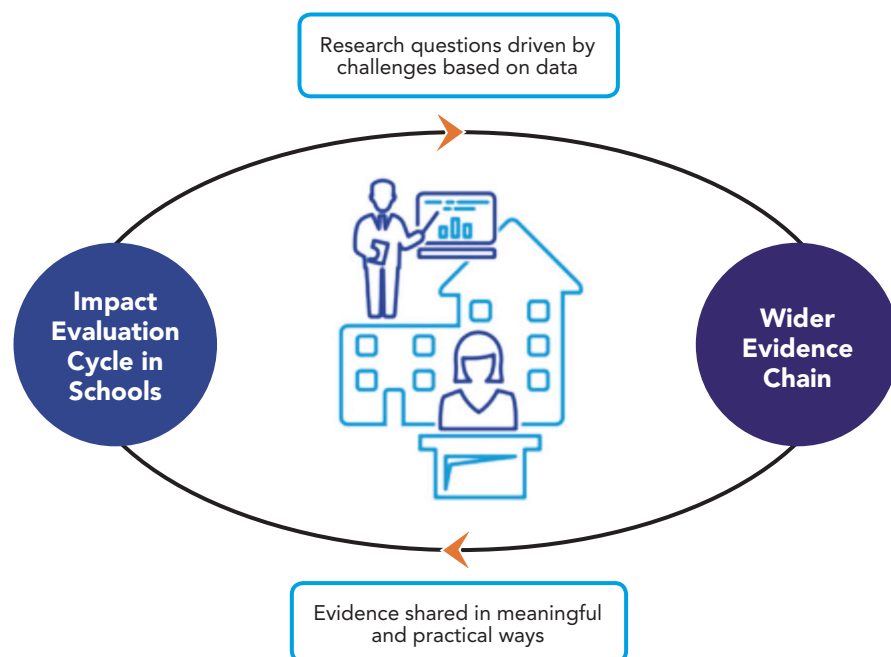
School leaders can ensure the approaches they choose in their school are informed by, and responsive to evidence. The ten stages involved in this process are depicted in Figure 3. The fact that each stage needs to be data- and evidence-based

does not replace professional judgement – it supports, enhances and improves this judgement.

- The **Impetus** must come from school level data and a thorough appreciation of local needs.
- School leaders need to have an **Awareness** of which approaches have good evidence of success and be able to distinguish them easily from the plethora of available information.
- **Analysis** of the evidence-based approaches entails understanding the conditions that have led to prior success and consideration of whether it is likely to provide similar benefit in the specific local context.
- Faithful **Adoption** and intelligent **Adaptation** of the program includes the use of implementation science to articulate the theory of change and manage the barriers to success (including staff training and measures to determine efficacy in that context).

School leaders can ensure the approaches they choose in their school are informed by, and responsive to evidence.

Figure 2. The evidence ecosystem



Source: Evidence for Learning, 2017b

- Implementation is conducted in a mini cycle of **Act, Evaluate and Adjust**. It ensures active learning to make immediate difference and generates new data and knowledge for sharing with the wider evidence chain.
- A deliberate decision to **Embed or Omit** the program based on the local evidence of impact; choosing what to stop (omit), is just as important as choosing what to keep (embed).

Wider evidence chain

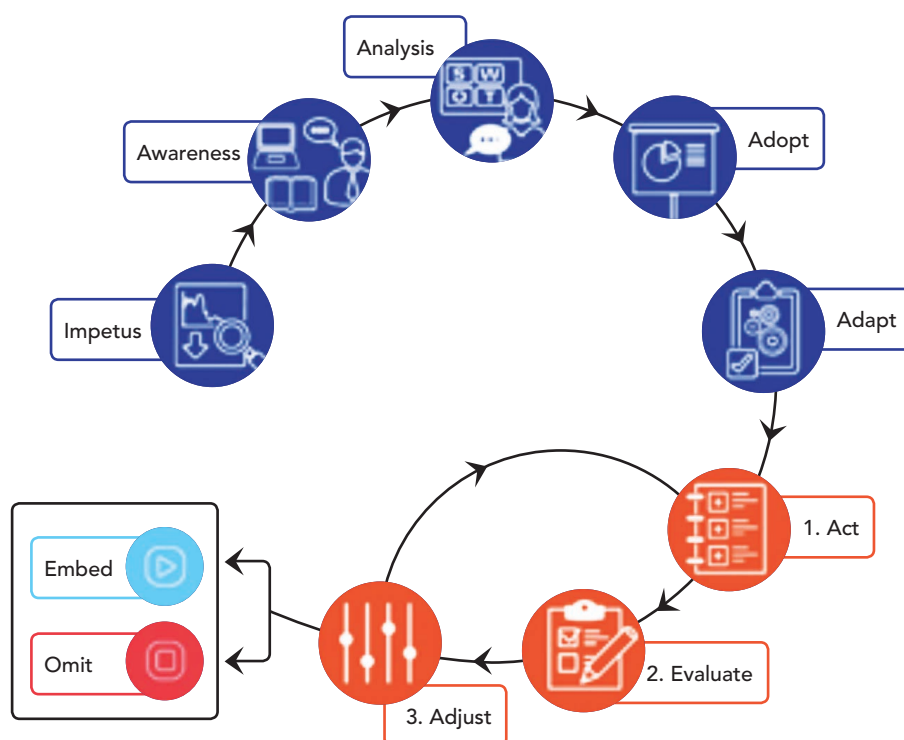
The wider evidence chain¹³ can be conceptualised as a cycle that begins with new knowledge generation in the **production** stage, based on the questions generated from data and evidence in the school system. To be useful and adopted, this evidence needs to be **synthesised** with similar research on the same topic and **transformed** into plain English with additional contextual information, such as costs and key success factors. Finally, it needs to be shared effectively

through authentic and engaged networks of frontline professionals. As they **engage** in an impact evaluation cycle in their school, teachers, school leaders and support staff must be supported in their **implementation** to realise the benefits of the evidence-informed change. See Figure 4.

Implications for researchers, practitioners and policy makers

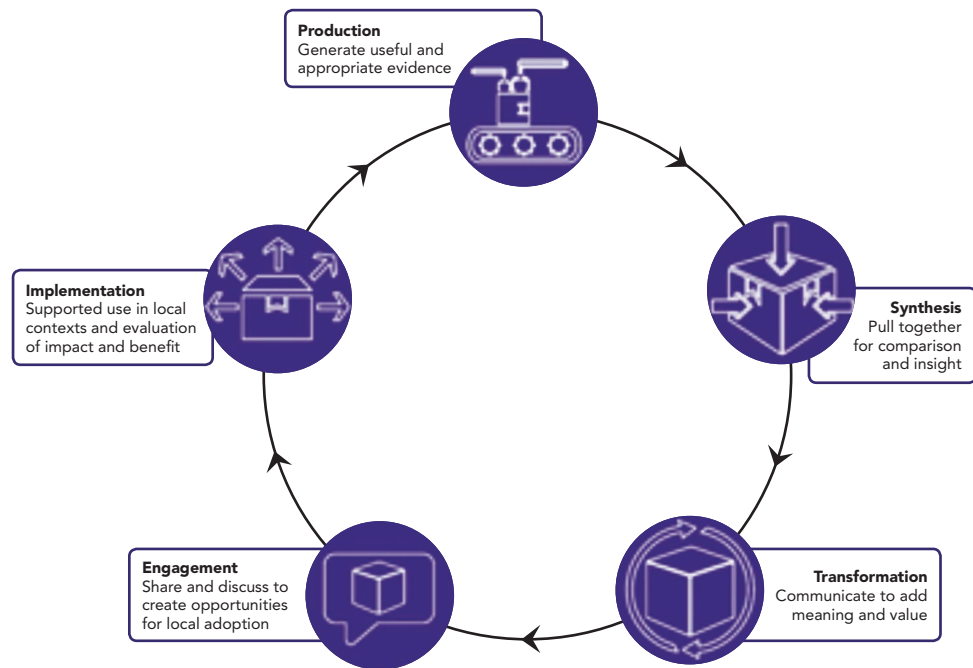
1. To effect change in practice, there is a need to **understand how research will be used by practitioners**. 'Creating evidence ecosystems ... requires coordinated efforts from a wide range of stakeholders [but] it is imperative that professionals drive these developments ... it is frontline professionals who ... should be at the heart of evidence – informed practice' (Sharples, 2013, p 24).
2. Placing frontline professionals and students at the heart of the work requires **updates to thinking about their role in research and evidence**. They cannot be regarded as passive recipients

Figure 3. Impact evaluation cycle in schools



Source: Evidence for Learning, 2017e

Figure 4. Wider evidence chain



Adapted from Sharples (2013)

Source: Evidence for Learning, 2017b

of knowledge but as active generators of new knowledge. ‘School leaders need to be continually working with their staff to evaluate the impact of all on student progression ... Schools need to become incubators of programs, evaluators of impact and experts at interpreting the effects of teachers and teaching on all students’ (Hattie, 2015, p 15).

3. The relationship between frontline professionals engaging in a cycle of impact evaluation in their schools and other contributors in the wider evidence chain must be one of **mutual dependence in a network of shared pursuit to improving educational outcomes**. ‘These networks would aim to inform educators as to what is more likely to work where, for whom, and under what conditions. Moreover, as educators used this knowledge, the knowledge itself would evolve and be further refined through its applications’ (Bryk, 2015, p 473).

Contributing to the ecosystem in Australia

Over the last three years Evidence for Learning has worked to advance a better evidence ecosystem for Australian schools. We think about our work in the following three areas:

1. Build – new independent evaluations of promising school based programs;
2. Share – promotion of a global evidence resource on approaches in schools with local research knowledge; and
3. Use – supporting in-school improvement cycles with resources and tools.

In doing this work we are learning from international experience and examples,¹⁴ but we are designed for the Australian context. School education in Australia is not a single system. Our country is a federation with eight states and two territories, each of which has the primary responsibility for school education,

working with a national government that has a crucial role to play. There are three school sectors (government, Catholic and independent), operating in all jurisdictions; and there are three national agencies (ACARA, AITSL and ESA) all playing critical roles at both national and state/territory levels. Evidence for Learning tries to avoid replication of anything these bodies already do and to add new value by tailoring our work to align with their strategies and priorities.

Whilst Evidence for Learning seeks to collaborate with existing actors, we are independent from all schools, systems, governments, academic institutions and program suppliers. Our influence only comes from how useful we are to systems and schools and, most importantly, to educators. We are transparent about our sources of funding and committed to publishing our work for free in open access forms. We are also transparent about the sources of evidence on which we rely and welcome critical challenge and debate. We hope that this approach generates trust in our work.

These principles informed our organisation's governance and advisory structure (Evidence for Learning, 2017c). We draw representatives from all levels, sectors and jurisdictions of the school system as well as representatives from the university, social-purpose and finance sectors.¹⁵ They help ensure that Evidence for Learning is responsive to the profession at the macro, school and practitioner level.

Building evidence through the Learning Impact Fund

The Learning Impact Fund pairs programs in schools with experienced independent evaluators to conduct rigorous research on their impact on learning (Evidence for Learning, 2017f). The Fund makes grants for evidence generation with quantitative measures of learning gain (randomised controlled trial designs where possible), as well as secondary measures and qualitative factors. It produces freely available, plain English reports for use by school leaders showing the months of learning gain, the cost to implement and the strength of the evidence. Evidence for Learning

Table 1: Randomised controlled trials

Program Name	Description	Number of Schools	System Partner	Independent Evaluator
Thinking Maths	Teacher professional learning intervention to make maths instruction deeper, more personalised and more engaging.	167 schools (7,367 students)	SA Department for Education	ACER
MiniLit	Small-group reading intervention for struggling Year 1 students, focusing on five keys: (1) phonemic awareness; (2) phonics; (3) fluency; (4) vocabulary; and (5) comprehension.	9 schools (237 students)	NSW DET	Murdoch Children's Research Institute Centre for Program Evaluation (MGSE)
QuickSmart Maths	A small-group tuition intervention to increase automaticity in maths and reduce cognitive load.	23 schools (293 students)	Sydney Catholic Education Office	University of Newcastle (Teachers and Teaching Research Centre)

Figure 5. The Teaching and Learning Toolkit

Sort by Name—	Average cost—	Evidence security—	Months' impact ▼
Feedback	\$ \$ \$ \$ \$	🔒 🔒 🔒 🔒 🔒	+8
Meta-cognition and self-regulation	\$ \$ \$ \$ \$	🔒 🔒 🔒 🔒 🔒	+8
Collaborative learning	\$ \$ \$ \$ \$	🔒 🔒 🔒 🔒 🔒	+5
Early years intervention	\$ \$ \$ \$ \$	🔒 🔒 🔒 🔒 🔒	+5
Homework (Secondary)	\$ \$ \$ \$ \$	🔒 🔒 🔒 🔒 🔒	+5

Average cost
The approximate cost of implementing an approach.

Evidence security
Based on the quantity and the methodological quality of the available evidence, and the reliability or consistency of impact estimates.

Months' impact
The additional months' progress you can expect students to make as a result of an approach being used.

Source: Evidence for Learning, 2017a

has commissioned three randomised controlled trials and one developmental study on programs in schools – see Table 1. The results from these trials will be made available starting in late July 2018.

Sharing evidence from the Teaching and Learning Toolkit

The Teaching and Learning Toolkit (the Toolkit – Figure 5) is a free online summary of global educational research, drawn from an international literature source of more than 13,000 studies (Education Endowment Foundation, 2017). The Toolkit synthesises international and Australian research, to estimate the average impact, cost and strength of the evidence for a wide range of teaching and learning approaches. The Toolkit is a live resource that is updated regularly as new studies are published. To meet our local Australian needs, the Toolkit was upgraded with an additional section focused on Australasian evidence, through a collaboration with the Melbourne Graduate School of Education at the University of Melbourne. Evidence for Learning is working with Departments of Education and national education

organisations (both non-profits and those that are federally funded) to map the 34 approaches within the Teaching and Learning Toolkit (Evidence for Learning, 2017a) to their policy frameworks. The aim of this work, which results in ‘tailored’ Toolkits, is to provide evidence within a localised context familiar to the frontline professional. As of early 2018, 15 tailored Toolkits have been developed that align with a broad set of policies and programs in Australia. Among them is a tailored Toolkit aligned with the School Excellence Framework by the NSW Department of Education and Training and the Centre for Education Statistics and Evaluation.

Using evidence through improvement cycles, practice guides, workshops and presentations

To engage the profession at the school and individual levels with the tailored Toolkits and the Impact Evaluation Cycle, Evidence for Learning conducts workshops with teachers and school leaders across the government, Catholic and independent sectors. These workshops are designed to

equip educators with practical knowledge of how to use evidence in their daily work and to generate pragmatic responses to their local challenges. The format and facilitation of the workshops have been adjusted to fit each audience. Most workshop sessions are spent with educators working hands on with the Toolkit, aligning a challenge in their school or within their practicum to an approach within the Toolkit, and sharing their findings specific to their context with the rest of the group. Workshops with the Toolkit have also been facilitated in various universities across Australia (eg, Melbourne Graduate School of Education, Queensland University of Technology and Murdoch University), and at conferences for undergraduate, post graduate education students, teachers and school leaders. Early this year a series of workshops was designed and facilitated collaboratively at Monash University, for their Initial Teacher Education and Masters Students.

Evidence for Learning, in partnership with the Australian Institute for Teaching and School Leadership (AITSL), developed a suite of feedback resources for schools (AITSL, 2017; Evidence for Learning, 2018). The free-to-access, evidence-based resources provide practical support in applying effective feedback practice in the classroom to enhance student learning outcomes.

Considerations for research, practice and policy for researchers

Education research has the ultimate aim of positively influencing students' learning. The closer that research is aligned to the needs of practitioners, the greater the chance the research produced will be used by them in their classrooms. This is why we put the practitioners at the heart of our

Some statements from practitioners

I spend a great deal of time collecting, reading and unpacking research around teaching practice and approaches with our teachers. The Toolkit is neat, concise and easy to use. It will save me countless hours in the way it lights the path directly to the most relevant and reliable research.

Frances Roberts, Head of Curriculum, Bounty Boulevard State School

We were very much driven by the research and the evidence-based approach that was provided by the Toolkit. All the hard work was done. We had lots of faith that this would work if we implemented it. I believe that it has.

Kath Perrier, Assistant Principal, Lourdes Hill College

This resource is outstanding and has given me a rich understanding of the impact of specific teaching and learning and supported decision-making of what best fits a particular student/cohort.

Trish Johnstone, Teacher, Kennington Primary School

Evidence Ecosystem. A critical challenge for researchers is ensuring their findings can be used practically by the profession. This should inform the design, execution and reporting of research findings.

When thinking about the selection of research projects, Evidence for Learning asserts that we should privilege questions that are relevant to practice, looking at practices and programs that are being used in schools and related settings

already. This has informed the selection of Evidence for Learning's first trials being school-based interventions, covering topics like a professional learning program for maths teaching, a reading program for early readers and a maths tutoring program to increase automaticity.¹⁶

When thinking about design we consider evidence quality. How much confidence can we have in the findings of the research? One aspect of evidence quality is the question of causation. How sure can we be that the change in approach is the reason that we have seen improved student learning outcomes? There are many ways to design research to address this question. We can use a quasi-experimental approach or we can use randomised controlled methodology (to name a couple). Where possible, randomised controlled methodology is preferred, as it can:

- eliminate selection bias and can generate a causal conclusion;
- avoid potentially misleading results from non-experimental work that has controlled inadequately for selection bias;
- provide a quick and digestible conclusion of program effectiveness that avoids lengthy caveats;
- incorporate results for future meta-analysis (Hutchison and Styles, 2010).

Another important question is whether there is sufficient 'power' to have confidence in the results? This is determined by sample size and the previous measured effects. It takes expertise to determine this.¹⁷ Once the sample size has been decided how can we work with system leaders and program developers to ensure that we recruit enough schools so we have sufficient power to detect the most reliable measure of effect? An evidence broker can help with this through being a bridge between the policy makers, researchers and practitioners.

How can the voice of the practitioner be represented accurately in the research? How can we make sure that teachers', leaders' and students' voices are included, so that we identify the 'active ingredients' (Sharples, 2013) and know what is important to get right in the subsequent implementation of the approach? Unsurprisingly for qualitative and mixed methods researchers, we must ask the practitioners and the students what was it that made it work for them. We need strongly skilled qualitative researchers working alongside our quantitative researchers, to ensure a coherent and compelling narrative at the end of the research. Copresenting at conferences with practitioners can also be a way to involve the profession authentically.

In the end, researchers need to make sure their results are presented in ways that are meaningful for practitioners. They must be easily digestible, so practitioners can turn evidence into action. This is described by Associate Professor Mark Rickinson (2016) as follows.

Practising educators ... want to know clearly what a research study found (findings) and what that means or could mean for them in their context (implications) – and (if positive findings) how they could get this result in their environment (implementation).

The primary function of any research output then is to communicate the main findings and implications of a research study for a particular audience or set of audiences. This may sound obvious or simplistic but very often such key messages are missing from or hidden within research outputs. This can be avoided by taking time to ensure clarity of findings, identification of implications – and considerations for implementation.

For practitioners

All practitioners across Australia, from the teaching principal at a small school, to school leaders, teachers and teaching assistants, want to be making the best possible gains for their students with the resources that they have. We know, in Australia, the variation between classrooms is greater than the variation between schools (OECD, 2016b, p 226). This indicates that in Australia there is not consistency in scaling up good practice within one classroom to become great practice across the school. Crucial to this conversation are two key questions: how can practitioners use evidence to inform their decision making; and how can practitioners share with their schools and more broadly their **practice-based** evidence.

When looking at evidence to inform decision making, it is important to consider several key pieces of information. What is the cost, what is the evidence security and what type of impact has this approach made to students' learning? The Toolkit provides these key pieces of information and can be helpful to inform decisions. Evidence for Learning is working with educators across Australia in workshops to help turn evidence into action.

Once a decision has been made, making a change towards an evidence-based approach can be complicated. There are many different factors to consider. Some of the key questions to consider when planning for a change are the following.

- Where are you going?
- How will you get there?
- What will tell you that you have arrived?

Evidence for Learning has developed an Education Action Plan to act as a scaffold to answer these questions, as mapped to our Impact Evaluation Cycle. We have been working with practitioners in workshops to help them embed evidence into their decision making. A key part of these sessions is to discuss 'what will tell you that you have arrived?' – that is, what is the practice-based evidence along the way, determining that you have made an impact?

In working with AITSL on the Feedback Implementation materials, Evidence for Learning developed a module called Evaluating Impact. Within this we detailed that evaluation (what will tell if you have arrived) needs to focus on the following two levels.

1. Measuring the impact by answering the question, Has there been an improvement in students' learning? (Hattie, 2015).
2. Identifying the active ingredients involved in the implementation of the approach that worked in your setting and how they worked (Sharples, 2013).

To answer these questions there needs to be both quantitative and qualitative data (Caldwell and Vaughan, 2012) to tell a coherent narrative to the school and the broader education community. A useful platform for educators to share their practice-based evidence is at the Australian researchED (Australian College of Educators, 2017), which started in the UK, and the Australian Council for Educational Research (ACER) Excellence in Professional Practice Conference (ACER, 2017). Practitioners can also publish their findings, so everyone can access their thinking and this can be spread, helping great practice become common practice.

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For policy makers

Over the three years that Evidence for Learning has been in existence, there has been a growing national discussion about the role of evidence in improving education policy and practice.

The Productivity Commission conducted an inquiry into the education evidence base in Australia in 2016 (reports released in May 2017),¹⁸ and recommended ‘a national policy effort to develop a high quality and relevant evidence base’ and identified five key activities, being

1. development of research priorities;
2. commissioning of high-quality education research;
3. adoption of rigorous research quality control processes;
4. development of researcher capacity; and
5. translation, communication and implementation of high-quality evidence.

Late in 2017 the Federal Government commissioned David Gonski AC to chair a *Review to Achieve Educational Excellence in Australian Schools*.¹⁹ The terms of reference included making recommendations on the most effective teaching and learning strategies and advising on ‘arrangements to ensure the ongoing identification and implementation of evidence based actions to grow and sustain improved student outcomes over time’.²⁰

Colloquially known as Gonski 2.0 the Review has been completed and its Report submitted to the government in March 2018. The Report was released just at the time of writing and includes a recommendation to ‘establish an independent institution to coordinate the strategic development of a national research and evidence base

through the sourcing and generating of research, and the synthesising and promotion of educational evidence that can be easily accessed and implemented to improve student outcomes.’²¹

Evidence for Learning participated in a submission to the Review.²² We recommended the Federal Government should establish and fund a national Education Evidence Broker, created through an open tender process, with an endowment of \$150m to be spent over 10 years to secure its long-term success and independence. We proposed nine features of the broker in order to be successful in Australia. They are as follows.

1. **Independent** – must be independent of governments, so there is no actual or perceived influence over the choice or conduct of evaluations and release of reports. This means its work can be relied upon and used by policy makers and educators in different jurisdictions and sectors. It must be able to choose the programs to evaluate within its governance structure. Independence is critical to the evidence being adopted by educators and leaders having confidence in the findings.
2. **Transparent** – must publish every finding in free and open forms for scrutiny; disclosure of results that show what has not worked are as important as results that show what has, to create an active discussion of failures. It must also show the funding behind the research, the trial design and methods and the data underpinning any findings. This encourages confidence in the institution and valuable review and critique of the work.
3. **High Quality** – must have an evidence standard that is rigorous, to ensure credibility and effective guidance.

This includes features of relevance to developmental stage, appropriate quantitative measures and a ‘causation focus’ with an appropriate control or comparison group.

4. **Long-term** – must have sufficient funding size and period to allow for continued focus on mission (instead of on funding protection) and to signal to all actors in the sector that the cultural change to evidence-informed practice is valued.
5. **Efficient** – must separate evaluators from the commissioning body so that it is not a monopoly (build capacity across the sector) and can be competitive and focused on the end needs of frontline professionals.
6. **Responsive** – must be aware of state, territory and national governments, Catholic and independent sector priorities and agenda to ensure research and mobilisation efforts are aligned and relevant to their strategic priorities.
7. **Useable** – must generate resources in plain English formats with easy-to-understand measures of impact and cost. These resources must also include specific implementation support, to ensure insights can become actions in schools. The resources will benefit if they are able to integrate easily into existing jurisdictional and sector frameworks and support tools.
8. **Able to leverage** – must be able to leverage government and system investment to encourage business and philanthropy to serve the national interest of a high-performing education system through further funding and support.
9. **Global** – must be integrated with international efforts to build a global education evidence base, to contribute

to and learn from global efforts in the field, including adapting promising international findings into an Australian context and sharing Australian research with the global evidence base.

Evidence for Learning proposed functions for the broker to provide a platform for evidence sharing and collaboration between the States and Territories. This included to coordinate a national research agenda and cross-jurisdiction trials; complement, not replace or duplicate, existing structures or services; commission (but not conduct itself) high-quality research trials and reports; and fund and support local organisations that work closely with schools.

In February 2018, the Federal Labor Opposition announced a policy to create and fund an ‘Evidence Institute for Schools’.²³ It would be independent of governments and charged with commissioning research, promoting evidence-based teaching practices and disseminating the findings to schools. Whilst more detail on the policy is required it does appear to meet many of the design features Evidence for Learning believes are required for a successful evidence broker in Australia.

The Federal Government has said that it supports the recommendation to establish a national evidence body (and the other recommendations of the Report) and will work with states, territories and non-government systems to implement them.²⁴ This means that both the Federal Government and Opposition have a commitment to placing evidence at the heart of school improvement. The release of the *Gonski 2.0* Report and the Government’s response to it will give greater clarity about how this agenda will develop in Australia over the coming years.

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The central premise of Evidence for Learning's work is the need to place the school, and leaders and teachers at the heart of a collective effort for improved outcomes.

Evidence for Learning hopes our journey continues to inform policy makers nationally, in the states and territories and across the sectors. What started as working principles initially have, through practical experience, become confirmed design features. Evidence for Learning strongly believes an independent, national and cross-sectoral body that is focused on building, sharing and encouraging the use of evidence to improve educational outcomes, and which works collaboratively with all stakeholders in our complex system, represents the best model for Australia and our learners.

Conclusion

This paper began with a reflection on the importance of curiosity to teaching and learning. It described how an attitude of inquiry and exploration is just as important for those seeking to make education systems better as it is to those whose focus is in the classroom. The paper has summarised an emerging 'evidence agenda' in Australian education and the role of a new national, non-profit organisation, Evidence for Learning, which has developed assets and is piloting activities in support of its mission of helping educators increase learning through better evidence. The paper has also set out implications and considerations for researchers, practitioners and policy makers.

The central premise of Evidence for Learning's work is the need to place the school, and leaders and teachers at the heart of a collective effort for improved outcomes. It is fitting, therefore, to conclude the paper with the story of another vital institution dedicated to children and young people.

Cincinnati Children's Hospital has been focusing on transformational improvement for more than a decade.²⁵ It is now ranked among the top three hospitals in the USA for six areas of practice in paediatric care. This was not always the case for the hospital, with lower than national average results in quality of care and results for patients. In charting a course towards excellent outcomes for children, then CEO Jim Anderson focused intensely on continuous improvement. He said: 'We may not be the best, but we can be the best at getting better, and then we will be the best' (Kotagal and Compton-Phillips, 2017)

The hospital summarises lessons learned on its journey as follows.

1. We have much to learn from the families we serve.
2. Large-scale change begins with transformational goals.
3. Build the organisation's capacity to execute improvement.
4. Transparency is critical.
5. Respect the science of improvement.²⁶

Dr Uma Kotagal, responsible for Safety, Quality, and Transformation at the hospital sums up how innate curiosity is strengthened to create continuously improving outcomes. It is just as relevant for those who strive for better learning outcomes in schools. She said

We're working with children who are always teaching us, so I would say our culture is collaborative, it is optimistic, it is transparent, it is aspirational, it's patient centered and it is scientific.

(Kotagal and Compton-Phillips, 2017)

Endnotes

1. Education plays a substantial role (accounting for 25–45 per cent) in economic immobility between generations (Centre for Education Statistics and Evaluation, 2015).
2. Australia's performance in international and national measures has been showing a steady decline or a flatlining over time (ACARA, 2017; Thomson, De Bortoli, and Underwood, 2017). The Programme for International Student Assessment (PISA) results for 2015 show decreased average achievement for Australian students since 2000 for literacy, 2003 for mathematical literacy and 2006 for scientific literacy (Thomson et al, 2017). The National Assessment Program Literacy and Numeracy (NAPLAN) results in reading have shown no significant improvement since 2008 for Years 9 and 7, although some statistically significant gains have been made in Years 3 and 5 (ACARA, 2017, p 258). Within the numeracy results there have been significant gains in Years 9 and 5, although none for students in Years 7 and 3 (ACARA, 2017, p 279). These results show that we are not making consistent gains for our students.
3. We have seen a growing divide in performance for those students surrounded by disadvantage (Lamb et al, 2015; OECD, 2016a; Perry, 2018; Thomson et al, 2017). The relationship between students' outcomes and their backgrounds is complicated, with a range of results from students that are from backgrounds of disadvantage (Thomson et al, 2017). In general terms, a student from a background that is socioeconomically disadvantaged is five times more likely to be a low performer (low performer is defined as those 15 year old students who score below Level 2 on the PISA mathematics, reading and science assessments) than students from an advantaged background (OECD, 2016a). Engagement of students can be a way to assist in the levelling of the influence of background on students' outcomes.
4. Engagement of students within the classroom increases learning; the converse, that disengagement decreases learning, is also true. Many students in Australia are disengaged (Angus et al, 2009; Gosset et al, 2017; Sullivan et al, 2014). Disengagement can be thought of as occurring at three levels (Goss et al, 2017, p 9), which are
 - passively disengaged behaviours: where a student is compliant but quietly disengaged from learning;
 - low-level disruptive behaviours: where a student is noisy, restless or interrupting others and disengaged in learning;
 - aggressive and anti-social behaviours: where a student is very uncooperative or fails to comply with classroom norms.

A study in Western Australia identified that over 40 per cent of students regularly displayed unproductive behaviours, with the majority of students displaying passively disengaged behaviours (Angus et al, 2009). Similarly, in South Australia, disengagement was a common occurrence within classrooms (Sullivan et al, 2014). Schools surrounded by disadvantage were seen to have higher rates of disengagement and low-level disruption (Goss et al, 2017). Teachers and school leaders are negatively affected by the disengagement of students that leads to ongoing stress and attrition amongst the educators.
5. An evidence broker can be thought of as having three roles: creating resources that distil and communicate evidence from research; convening partnerships between researchers and practitioners; and supporting practitioners to engage with evidence and test its impact locally (Bush, 2017).
6. www.socialventures.com.au/.
7. Provided \$6m in 2015 as part of a \$50m commitment to schooling – see www.commbank.com.au/about-us/news/media-releases/2015/commonwealth-bank-50m-investment-boosts-teachers-and-schools.html.
8. educationendowmentfoundation.org.uk/.
9. en.oxforddictionaries.com/definition/evidence.
10. 'Verifiable by observation or experience rather than theory or pure logic' – Oxford English Dictionary.
11. Others are logical (reasoning-based) and anecdotal (personal experience and observation).
12. This is sometimes termed 'realist evaluation' – see www.betterevaluation.org/en/approach/realist_evaluation.
13. This work builds on the model in Jonathan Sharples' (2013) *Evidence for the Frontline*. Accessed at www.alliance4usefulevidence.org/assets/EVIDENCE-FOR-THE-FRONTLINE-FINAL-5-June-2013.pdf.
14. Evidence for Learning is the Australian licensee of the materials created by the UK's Education Endowment

Foundation and we are supported in our local work by their team. We draw on insights from the USA's Institute for Educational Sciences which provides 'scientific evidence on which to ground education practice and policy and to share this information in formats that are useful and accessible to educators, parents, policymakers, researchers, and the public.' Another notable organisation is the Knowledge Network in Applied Education Research (KNAER), which operates through an agreement between the University of Toronto, Western University and the Ontario Ministry of Education. The focus of KNAER is to 'build, advance, and apply robust evidence of effective practices through promoting research use'.

15. Evidence for Learning has established three governance and advisory bodies. 1) Steering Group, which acts as a corporate board ensuring we stay true to our mission, deliver on our approved annual business plan and meet our legal and financial obligations; 2) Expert Reference Council, which is made up of independent experts in education and school improvement. It provides sector input and guidance on priority areas and engagement; 3) Research Use and Evaluation Committee, which is made up of technical and practitioner experts in education research and evaluation. It provides guidance on the work of the Learning Impact Fund and research engagement in schools more generally.
16. evidenceforlearning.org.au/lif/our-projects/.
17. www.huffingtonpost.com/robert-e-slavin/what-is-a-large-effect-size_b_9426372.html.
18. www.pc.gov.au/inquiries/completed/education-evidence/report.
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21. docs.education.gov.au/documents/through-growth-achievement-report-review-achieve-educational-excellence-australian-schools.
22. Full copy of the submission is at evidenceforlearning.org.au/assets/News/SVA-Response-to-Review-to-Achieve-Educational-Excellence-in-Australian-Schools.pdf.
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Additional reading

Although not cited explicitly in the text, the following item was used in preparing this paper, and may be of interest to the reader.

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L-R: Matthew Deeble and Tanya Vaughan

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About the Paper

The authors explain the role of their non-profit organisation, Evidence for Learning, which has developed over the last three years as a pilot to inform Australia's education system and help educators increase learning through better evidence. The central premise of their work is to place the school, and leaders and teachers at the heart of a collective effort for improved outcomes. The authors argue that an independent, national and cross-sectoral body, which is focused on building, sharing and encouraging the use of evidence to improve educational outcomes, and which works collaboratively with all stakeholders in our complex system, represents the best model for Australia and our learners.

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